

Mathematical pendulum

Homework 3

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Contents

1. Mathematical model of the pendulum	1
1.1. Converting to first-order system	1
1.2. DOPRI5	2
1.3. Evaluating and comparing the mathematical and harmonic pendulum	2

1. Mathematical model of the pendulum

The angle movement of the mathematical pendulum is described with the equation:

$$\theta''(t) + \frac{g}{l} \sin(\theta(t)) = 0, \quad \theta(0) = \theta_0, \quad \theta'(0) = \omega_0, \quad (1)$$

where l is the string length and g is the gravitational acceleration.

To evaluate the function, we will use the DOPRI5 method, which is a version of the Runge-Kutta methods. But Runge-Kutta methods work on first order differential equations, while our equation is of the second order.

1.1. Converting to first-order system

Let

$$\omega(t) = \theta'(t) \quad (2)$$

be the rate at which the angle changes (angular velocity).

Then

$$w'(t) = \ddot{\theta} = -\frac{g}{l} \sin(\theta) \quad (3)$$

The result is the following first-order system

$$\begin{aligned} \theta'(t) &= \omega(t), \\ \omega'(t) &= -\frac{g}{l} \sin(\omega(t)). \end{aligned} \quad (4)$$

Combine the two variables θ, ω into one vector

$$u(t) = \begin{pmatrix} \theta(t) \\ \omega(t) \end{pmatrix}. \quad (5)$$

Runge-Kutta methods are designed for equations that look like

$$u'(t) = f(t, u). \quad (6)$$

This system can also be represented in the following vector form

$$u'(t) = f(t, u, p) = \begin{pmatrix} \omega \\ -\frac{g}{l} \sin(\theta) \end{pmatrix}, \quad u(0) = \begin{pmatrix} \theta_0 \\ \omega_0 \end{pmatrix}. \quad (7)$$

where the parameter parameters combined into $p = (g, l)$.

1.2. DOPRI5

Runge-Kutta methods, unlike e.g. Euler method, compute several intermediate slopes. R-K methods, when approximation between points $x_n \rightarrow x_{n+1}$ takes into account multiple derivatives between the two points, and creating a weighted sum (average):

$$\beta_1 f(x_n, y_n) + \beta_2 f(x_n + c_2 h, y_n + d_2 k) + \beta_3 f(x_n + c_3 h, y_n + d_3 k) + \dots \quad (8)$$

In general a single slope is calculated as

$$k_i = f\left(t_n + c_i h, y_n + h \sum_{j < i} a_{ij} k_j\right), \quad (9)$$

then the slopes are combined into a single weighted sum

$$y_{n+1} = y_n + h \sum_i b_i k_i, \quad (10)$$

where b_i are the weights.

The parameters c_i and a_i are often stored together in a Butcher tableau.

DOPRI5 is an embedded seven stage Runge-Kutta method that calculates two approximations at the same time. One approximation is of order 5 and the other is of order 4. The difference between the approximation gives us an estimate of the numerical error:

$$e = y_{n+1}^5 - y_{n+1}^4. \quad (11)$$

If the difference is too big, the step was too large and be repeated using a smaller h . Thus DOPRI5 can adapt its time step.

The Butcher tableau for the DOPRI5 method can be found here https://en.wikipedia.org/wiki/Dormand%E2%80%93Prince_method#Butcher_tableau.

1.3. Evaluating and comparing the mathematical and harmonic pendulum

The harmonic pendulum formula can be directly solved using the following stuff:

To track the period and to interpolate for plotting, we will use the Hermitic interpolation and zero finding as in Vaje 16.